

Robot as a Peer in Learning: How the Robot’s Role Changes Help-Seeking and Comfort

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1 Introduction

Schools and universities now use social robots in classrooms. Most of the time, robots play one of two roles. The first role is the tutor. A tutor robot knows the correct answer and gives it to the student. The second role is the teacher. A teacher robot leads the whole lesson from the front of the class.

Belpaeme et al. [2018] and other researchers wrote a big review paper about this topic. They looked at 101 different studies and 309 results about robots in education. They found that robot tutors really do help students. When a robot acted as a tutor, students learned better, with a strong effect on learning (0.70) and a strong effect on feelings about learning (0.59). These numbers are close to the number for human one-to-one tutoring (0.79), but a little lower. The review says that one big reason for this success is that the robot is physically present in the room. People listen more, try harder, and learn faster when a real robot is with them, compared to a video or a computer character [Belpaeme et al., 2018].

The review also talks about a third role that is not studied very much: the robot as a peer. A peer robot is not above the student. It is next to the student, like a friend. The peer robot tries to solve problems together with the student, instead of just giving answers. The idea is that a peer feels less scary than a tutor or a teacher. Working with a peer may also be better than working with a robot that judges you all the time. Belpaeme et al. [2018]’s review gives some early proof for this idea. Kanda et al. [2004] found that children paid attention for longer, answered faster, and gave more correct answers when they worked with a peer-style robot, compared to a tutor-style robot that looked the same. Zaga et al. [2015] also compared peer and tutor robots and looked at how engaged children were in the task.

But there is a problem: there is very little research on the peer robot role. Out of the 309 results in the big review, only 9% used a robot as a peer or as a beginner. Most of the results (48%) used the tutor role, and many others (38%) used the teacher role [Belpaeme et al., 2018]. Also, almost all of the peer-robot research was done with children, not adults. The review itself says clearly that there is not enough evidence for the peer effect yet. This gap between how promising the peer idea sounds and how little we actually know about it, especially for adults, is the problem that our small study tries to look at.

Our research question comes directly from Belpaeme et al. [2018]’s review. The question is: does a robot’s role as a peer, instead of a tutor, change how often adult students

ask for help, and how comfortable they feel while learning? We wrote two hypotheses to test this question:

- **H1:** Students working with a peer robot will ask for help more often than students working with a tutor robot.
- **H2:** Students in the peer group will feel more comfortable (and less stressed) than students in the tutor group, even if their test scores are similar.

Both hypotheses come from the same idea in Belpaeme et al. [2018]’s review: if the peer robot really makes it feel safer to ask for help, we should see this in two ways. First, in behaviour more help requests. Second, in feelings higher comfort scores. But we should not see a big jump in quiz scores, because the idea is about willingness to ask for help, not about giving away the answers. As we explain in the Results section (5), this last part ‘no jump in scores’ became the most difficult part of our whole study. It is the finding we most want to report honestly, not hide.

2 Related Work

Apart from the Belpaeme et al. [2018] review, a few other studies helped us plan our quiz. It is important to explain exactly what we took from each study, instead of just listing names.

Zaga et al. [2015] studied how a robot’s social character peer or tutor changes how much children engage in a discovery-learning task. Their idea was that ‘social character’ comes from what the robot says, not from how the robot looks. We copied this idea directly: our peer and tutor groups used the exact same chat interface and the same quiz questions. The only thing that changed was the words used in the hints and in the welcome message. If the robot looked different in each group too, we could not be sure if any difference came from the role or from the appearance.

Kanda et al. [2004] used a robot called Robovie in a real elementary school. This was one of the first robots that worked fully on its own in a school. They found that a peer-style robot made children pay attention for longer and give more correct answers, compared to an identical tutor-style robot. This is one of the studies Belpaeme et al. [2018]’s review uses to say that peer robots ‘might’ feel less scary. It is also where we got the idea that a peer robot could change accuracy, not just comfort. Something similar happened in our own data too, but for a different reason, which we explain in Section 5.

Salomons et al. [2022] helped us decide to test this with university students instead of children. Their study compared a peer robot and a tutor robot teaching adults how to build electronic circuits. They found no big difference in the total number of skills learned between the two groups. But students who knew very little about the topic before the study learned more with the peer robot [Salomons et al., 2022]. Also, students rated the peer robot as friendlier, more social, smarter, and more respectful, no matter how much they already knew [Salomons et al., 2022]. This taught us two things. First, the peer-versus-tutor question is not only about children - adults show effects too. Second, when a peer robot does better than a tutor robot, it is usually true only for some students, not for everyone. This made us careful about trusting a simple ‘the peer group scored higher’

result. It also pushed us to check our help-request data very closely, which turned out to matter a lot (see Section 5.4).

Diyas et al. [2016] ran a shorter study that compared a peer robot and a teacher robot while teaching programming. We did not use their exact results much, but their method keeping the lesson the same and only changing how the robot presents itself supports the same idea as Zaga et al. [2015]’s study. This gave us confidence to use one shared quiz for both of our groups, instead of writing two separate quizzes.

Chase et al. [2009] wrote about something called the protégé effect. This is the idea that people try harder and learn more when they believe they are teaching or helping someone else, instead of being taught themselves. Our study is not exactly the same, because our robot is not a student being taught by the participant. But this idea explains why a peer robot that says things like ‘we will figure this out together’ might change how much effort and comfort a person feels. This is why our peer robot’s hints (Section 3.1) were written using words like ‘we’ and ‘us’, instead of ‘you’.

Dillenbourg [1999] wrote an older, more theoretical paper about what ‘collaborative learning’ really means. We used this mainly to check our hint wording. Dillenbourg [1999]’s main point is that collaboration is not just about being in the same place it is about equal roles and shared goals. This is why we tried, not always successfully, to make the peer robot’s hints sound like joint thinking, such as ‘I’m not sure about this one either’, instead of secretly giving away the answer.

Bainbridge et al. [2011] is a paper we return to in our discussion (Section 6). It talks about the limits of running a study through a laptop screen instead of using a real physical robot. They found that robots that are physically present create stronger effects than the same robot shown on a video [Bainbridge et al., 2011]. This is also one of the main arguments in Belpaeme et al. [2018]’s review. It is a real limitation in our own study, because we did not have the equipment to use a physical robot.

3 System Description

Our system is a small website built with Flask (a Python web tool). The main file is called ‘app.py’, and it shows one page (‘templates/index.html’) that looks like a chat with a robot.

In the current version of the code, the participant’s group is chosen by the server before the page even loads. The code picks randomly between ‘peer’ and ‘tutor’ using the line: ‘condition = random.choice(['peer', 'tutor'])’. The starting screen no longer lets the participant click a button to choose their own role. Instead, it simply tells them which role was picked for them, for example: ‘Your robot companion has been randomly assigned: Peer (Study Buddy)’ or ‘Tutor (Instructor)’.

This point is important, because it fixes a real problem in an older version of the system. In that older version, participants could click a button and pick their own group — ‘Peer’ or ‘Tutor’. This is called self-selection, and it is not the same as random assignment. We explain more about this problem in Section 4.2. It matters a lot, because the data used in this report (Section 5) was collected using that older, self-selecting version of the system, before the fix was made.

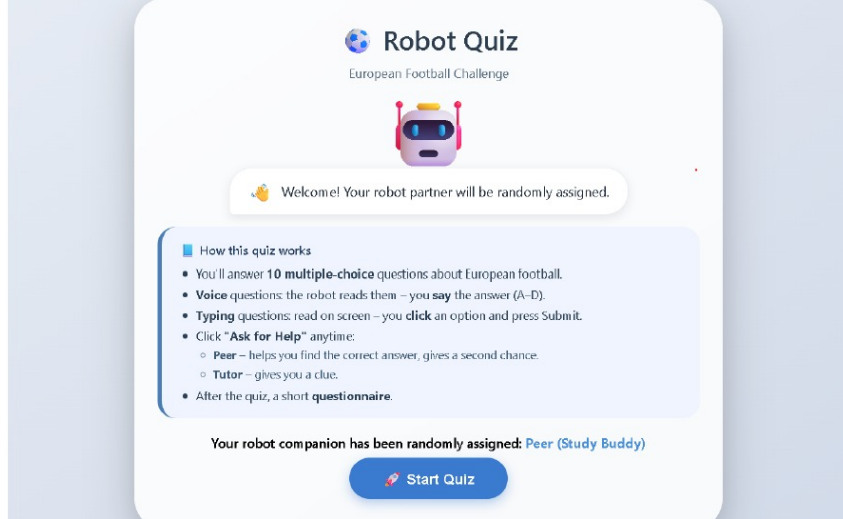


Figure 1: The starting screen. It shows the quiz title, short instructions, the sentence ‘Your robot partner will be randomly assigned’, and a Start Quiz button.

After the participant clicks ‘Start Quiz’, the robot does not go straight to question one. First, it speaks a longer, role-specific message out loud, using the browser’s speech feature. While this message is playing, the participant cannot do anything else they cannot answer questions or ask for help until the robot finishes talking. Only after that does the first question appear. We explain this part in full detail in Section 3.2, because the two messages are different enough between groups that we treat this as part of the study design, not just a small decoration.

After the message, the ten questions appear one at a time, each with four multiple-choice answers. Depending on the question, the participant answers either by speaking (using speech recognition, with a backup option if the browser does not support it) or by typing or clicking.

The quiz questions are about European football for example, which club has won the most Champions League titles, or who scored the most goals at the 2022 World Cup. There are two sets of ten questions (Set A and Set B), and participants get one set at random. We chose football instead of a school subject like maths or science, mainly to keep the quiz fair and easy for everyone. Some people know a lot about football, some people know very little, but the difference is not related to their university studies. This also kept the twenty-minute session light and relaxed, which was important for our comfort measurement.

Both question sets have similar difficulty: about one very easy question, several medium questions that football fans would know, and a couple of hard, less-known questions. This was done so that any score difference between groups would come from the robot’s role, not from an easier or harder question set. Each question also has a short factual clue (for example, ‘They have won it 14 times, more than any other club’). This clue is only ever shown to participants in the tutor group.

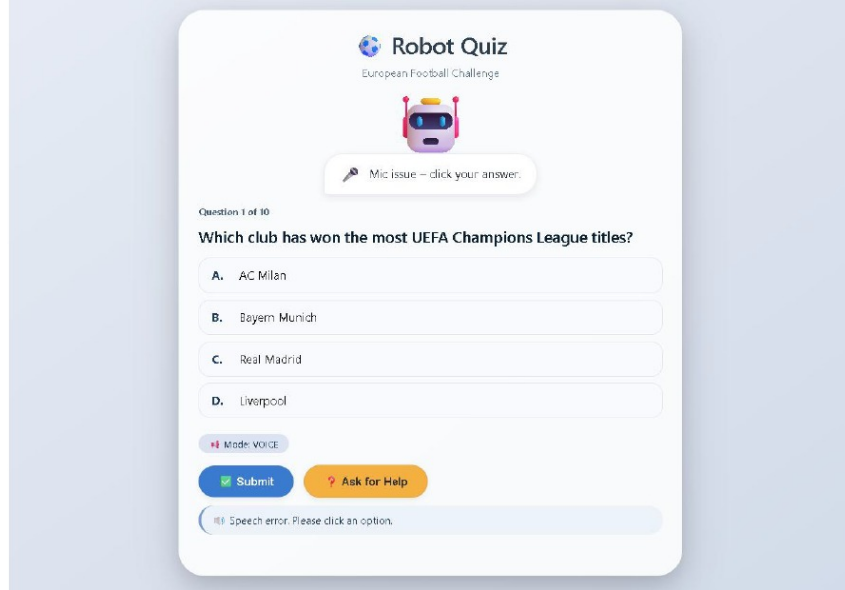


Figure 2: A question in progress. It shows the question text, four answer choices, the robot’s speech bubble, and the ‘Ask for Help’ button.

3.1 The Hint System

This part of the system is the main way we tried to show the difference between a peer robot and a tutor robot. It is also, as we explain later in Section 5.4, the main reason some of our results were affected. When a participant clicks ‘Ask for Help’, the server does two very different things, depending on the group:

In the peer group, the server picks one random wrong answer out of the remaining choices, and gives a friendly message telling the participant to remove it. The messages are written using words like ‘we’ and ‘let’s’, for example: "I’ve got a feeling {wrong_option} might be a trap. Let’s leave it out – you’re doing great!" The wrong answer is then visually crossed out on the screen. If the participant keeps clicking the help button, the system keeps removing wrong answers, one at a time, until only the correct answer is left.

In the tutor group, the server does not touch the answer choices at all. Instead, it shows one of four short factual clues, for example: "Here’s a clue: {clue}." This means the participant still has to think and reason using the clue, instead of just being handed a smaller list of choices.

There is one more difference. In the peer group, if a participant gives a wrong answer, they get a second try before the correct answer is shown. In the tutor group, there is no second try the participant is told immediately if the answer is right or wrong. Both of these choices were meant to make the peer robot feel kinder and more supportive. But looking back, we now see that these choices also made the peer group’s quiz mechanically easier, no matter how ‘peer-like’ the robot actually felt. We explain this clearly here, in the system description, because it is important background for understanding our results honestly, not something we want to hide until the very end.

3.2 The Message Before the Quiz

Both groups now hear a spoken message right after clicking 'Start Quiz', before the first question appears. The goal was to extend the peer/tutor difference to the very start of the session, similar to how a real tutor or study partner would introduce themselves before starting, instead of just showing a plain instruction screen.

However, the two messages are not equal in length or content, and we think this is worth explaining clearly, instead of hiding it. This is a new source of difference between the two groups, on top of the differences already described in Section 3.1.

The tutor message is about 75 words long and is written in a formal, serious tone: 'Good day, student. Today we will explore the fascinating history of European football. . . I expect you to think carefully.' It talks about the quiz like a graded test, and it mentions only once that a clue is available if needed.

The peer message is about 110 words long about 45% longer than the tutor message and it is written in a casual, friendly tone: 'Hey! So you're into football? Me too! we'll figure it out together.' The peer message also tells participants, before the quiz even starts, that they will get a second chance if they answer wrong the first time. This means peer-group participants already know they have a safety net before they begin, while tutor-group participants are never told about anything similar in advance. This could easily make peer participants feel more relaxed from the very start, for reasons that have nothing to do with the robot feeling like a 'peer'. We treat this as an extra limitation in Section 6.1, instead of mixing it silently into our comfort result.

3.3 A Hint System We Tried and Gave Up On

Before we built the simple template system above, we first tried to generate the peer robot's hints using a small language model (called DistilGPT-2) running on our own computer. The idea was to create more varied, natural-sounding hints instead of using ten fixed sentences.

This did not work well enough, so we stopped using it. First, it was slow - each hint took about three to five seconds to generate, which was long enough that participants noticed the delay and mentioned it. Second, the generated text was often unreliable. Sometimes it made up facts, or it gave hints that did not match the correct answer. This is clearly not acceptable for something meant to help a participant find the right answer. Third, running the model used a lot of computer memory and processing power, which put a strain on the laptops we used during sessions. Finally, since the model's hints were different every time, it would have been very hard to study them properly afterward - we would have had to read and label every single hint by hand, instead of simply recording which fixed template was used.

The system we ended up using a set of pre-written hint templates, chosen randomly, with details filled in automatically is a simple idea, but it solved these problems well. The response time became less than a hundred milliseconds. The hints were always factually correct, because we wrote them ourselves in advance. And every hint given could be recorded and traced back to its exact template. This made it possible to look closely at what happened during each participant's help-seeking, after the study was finished. This is exactly what allowed us to notice the option-elimination pattern described later in Section 5.4.

4 Methods

4.1 Participants

Fourteen people took part in our study ($n = 14$). Seven were placed in the peer group, and seven were placed in the tutor group. All participants were university-connected adults, aged between 23 and 26. The group of participants was mostly male — 12 out of 14 people. We want to say clearly that this is a limitation of our study, not something to ignore. With only two female participants in total, we cannot draw any real conclusions about gender in this study.

We also want to be honest that our sample size is smaller than planned. Our ethics application said we would aim for 20 to 24 participants (10 to 12 per group). Our final number, 14, is smaller than that. At the same time, it is larger than the original pilot test, which had only 6 participants (3 per group), as described in our early project slides.

4.2 Study Design and Group Assignment

Our study used a between-subjects design. This means each participant only experienced one condition peer or tutor not both. We chose this design to avoid carry-over effects, which could happen if the same participant experienced a friendly robot and then an authority-style robot right after each other [Belpaeme et al., 2018].

It is important to separate two different things here: how the system assigns groups today, and how it assigned groups when the data used in Section 5 was actually collected. These are two different versions of the code, and this difference matters.

The current version of ‘app.py’ assigns the group randomly on the server, before the page loads (`condition = random.choice(['peer', 'tutor'])`). The starting screen simply tells the participant which role they were given, instead of letting them choose. This matches what our ethics application and our early project slides describe: ‘random assignment to one condition (peer or tutor) by coin flip.’

However, the 14 participants whose data is analysed in Section 5 took part using an earlier version of the system. In that earlier version, participants picked their own group by clicking either a ‘Peer’ button or a ‘Tutor’ button on the starting screen. This is a self-selection method, not a random one, even though our planning documents describe random assignment. We only discovered this difference while comparing our code to our planning documents while writing this report. We want to be clear: the fix that exists in the current code does not apply backward to the data we already collected. The results in Section 5 still have this self-selection problem, even though a new participant using today’s version of the system would not have this problem.

Any future data collected using the current version of the system would not have this particular self-selection problem. However, the hint-system differences described in Section 3.1, and the message-length difference described in Section 3.2, are still present in the current code and would need to be fixed separately.

Because participants chose their own group in the data we studied, there is a risk of a self-selection bias. For example, people who are naturally more comfortable asking for help, or people who are generally more relaxed, may have been more likely to click ‘Peer’. We discuss what this means for H1 and H2 in Section 6.

4.3 Materials and Procedure

Our materials included the chat-style website described in Section 3, the ten-question quiz (Set A or Set B, given randomly), and a short questionnaire at the end. This questionnaire asked about comfort (using a 1–7 scale), one question to check if the robot’s role felt as intended, and two questions about age and gender.

Each session followed the same steps. Participants first read a short information sheet, and in sessions that followed the ethics protocol, they signed a consent form. They then picked a group (using the older, self-selecting version of the system), heard the robot’s welcome message for their group, and were told they could ask the robot for help at any time. They then answered the ten questions one at a time. Every help request, every retry, and every final answer was automatically saved into two files: ‘experiment data.csv’ and ‘help log.csv’. After the tenth question, participants filled out the short questionnaire and were thanked for taking part. Each session lasted about 20 minutes, as planned. All sessions were done remotely, using a locally hosted server that participants could access during their session.

4.4 What We Measured

We measured three main things, plus one check question:

Table 1: Study measures and their purpose

Variable	How we measure it	Why we measure it
DV1 Help requests	Count of "Ask for Help" button presses during the quiz, logged automatically for every question	Main test for H1
DV2 Comfort rating	Scale from 1 to 7, asked in questionnaire after the task	Main test for H2
DV3 Task score	Number of questions answered correctly, from 0 to 10	Control check, something we check but do not expect to change
Manipulation check	"Did the robot act more like a classmate or more like a teacher?"	To confirm that the role was perceived like we planned

One small but important note: our ethics application described the task-score variable as ‘percentage correct’. But the system, as actually built, records and studies a simple count out of ten instead. This does not change the statistics, because it is just a different scale of the same number. We mention it only so our planning documents and our real measurements stay consistent with each other.

4.5 How We Analysed the Data

We compared all three variables between the two groups using independent-samples t-tests, a common statistical test, with a significance level of $\alpha = .05$. Because our hy-

potheses had a clear direction (peer group higher than tutor group), we used one-tailed tests for H1 (help requests) and H2 (comfort). For the score variable, we used a two-tailed test, because we did not predict a direction for it - in fact, following Belpaeme et al. [2018]’s idea about the peer role, we expected scores to be similar in both groups. We also calculated an effect size (Cohen’s d) for every comparison.

In addition, we went back and looked closely at the manipulation-check answers and the raw entries in ‘help log.csv’. This closer look is what allowed us to discover the option-elimination pattern described in Section 5.4 - something that a simple t-test using only the total help-count would never have shown us.

5 Results

Table 2 shows the basic statistics for all three variables, for each group (7 participants per group).

Table 2: Descriptive statistics by condition

Condition	n	Score (M $\pm$ SD)	Help Count (M $\pm$ SD)	Comfort (M $\pm$ SD)
Peer	7	7.29 $\pm$ 2.36	12.14 $\pm$ 11.84	5.71 $\pm$ 0.95
Tutor	7	3.57 $\pm$ 1.72	6.00 $\pm$ 3.21	3.57 $\pm$ 0.79

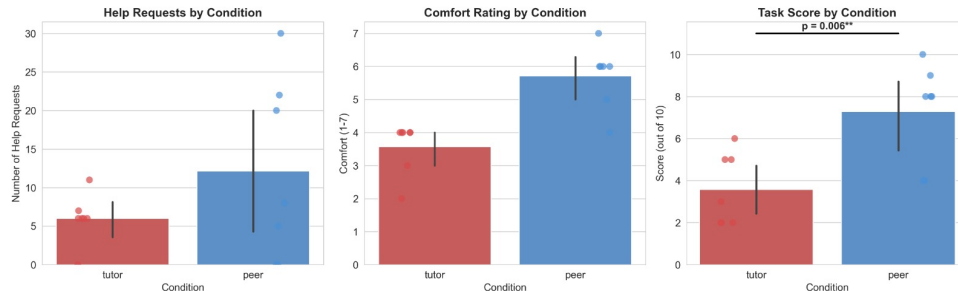


Figure 3: A chart (made by our analysis script) shows help requests, comfort scores, and task scores for both groups, with each participant’s individual result shown as a dot.

5.1 H1 – Asking for Help

On average, peer-group participants asked for help more often ($M = 12.14$, $SD = 11.84$) than tutor-group participants ($M = 6.00$, $SD = 3.21$). This is roughly twice as many requests. However, the one-tailed t-test on this result was not statistically significant: $t(12) = 1.325$, $p = .105$. The effect size was medium-to-large ($d = 0.71$). This means H1 is not supported at the normal significance level of .05.

The reason the result is not significant, even though the average difference looks large, is easy to see in Figure 3. The peer group’s help-request numbers are very spread out. Some participants asked zero times, while one single participant asked 30 times. The tutor group, on the other hand, is much more tightly grouped, from 0 to 11 requests. With only seven participants in each group, one unusual participant is enough to strongly

change both the average and the variation. This is exactly what happened here, and it is why we went back to look at ‘help log.csv’ directly, instead of trusting the summary numbers alone (see Section 5.4).

5.2 H2 – Comfort

Peer-group participants reported clearly higher comfort ($M = 5.71$, $SD = 0.95$, on a 1–7 scale) than tutor-group participants ($M = 3.57$, $SD = 0.79$). This difference was statistically significant: $t(12) = 4.593$, $p < .001$, one-tailed, with a very large effect size ($d = 2.46$). On the surface, this is the clearest result in our whole study, and it fits best with the peer-robot idea from Belpaeme et al. [2018]’s review. Participants who worked with the ‘study buddy’ robot rated their experience as noticeably more comfortable than those who worked with the ‘instructor’ robot. We explain in Section 5.4 and Section 6 why we do not think this result can be simply explained by the peer framing alone.

5.3 Control Variable – Task Score

This is where our study runs into its main problem. We expected the task-score variable to show no real difference between groups, following the logic behind H2 (‘even if task scores are similar’). The peer framing was only supposed to change how comfortable participants felt about asking for help, not how many questions they answered correctly.

Instead, peer-group participants scored much higher ($M = 7.29$ out of 10, $SD = 2.36$) than tutor-group participants ($M = 3.57$, $SD = 1.72$). This difference was statistically significant: $t(12) = 3.366$, $p = .006$, two-tailed, with a large effect size ($d = 1.80$). This is also shown directly on Figure 3.

This is a large effect on a variable that was supposed to show no real difference. It also makes the comfort result in Section 5.2 harder to trust cleanly, because success and comfort are connected. Scoring almost twice as well on the quiz is, by itself, a perfectly good reason to feel more comfortable afterward - it does not have to be about how ‘peer-like’ the robot felt. Section 5.4 explains why we believe this score gap is mostly a technical side-effect of how the two hint systems were built (Section 3.1), rather than a real psychological or social effect of the robot’s role.

5.4 Why the Score and Help-Count Results Look Like This

Looking at the raw entries in ‘help log.csv’ makes the score result much easier to explain, but it also makes the H1 story more complicated. One participant in the peer group (participant number 2007) asked for help exactly three times on almost every question which is the maximum useful number of requests, because three wrong options can be removed one at a time, leaving only the correct answer. The logged hint messages for this participant follow the same pattern on nearly every question. For example, on question 1, three different wrong answers were removed in three requests, just a few seconds apart. This participant finished with a perfect score of 10 out of 10, and 30 logged help requests the highest score and by far the highest help count in the entire study. This one participant is a major reason why the peer group’s average help count and average score both look inflated.

This is not really 'asking for help' in the way our study meant to measure it. Because the peer group's hint system directly removes a wrong answer, instead of giving an indirect clue, and because there is no limit on how many times a participant can ask, the system allows a participant to simply guess-and-eliminate their way to the correct answer, three requests at a time. The tutor group's hint system is different - it always shows the same short text clue, no matter how many times it is requested, so asking again and again cannot mechanically raise a tutor-group participant's chance of getting the right answer, the way it can in the peer group.

Once we take this into account, the higher average score in the peer group looks much less like proof that the peer framing helps learning, and much more like a direct, mechanical result of the peer group's help system simply giving away more information per request. This is combined with the extra retry chance the peer group gets after a first wrong answer (Section 3.1), which the tutor group does not get.

For H1 specifically, this also means our simple help-count comparison was actually measuring two different things mixed together: participants who asked for help because they were genuinely unsure and wanted support (which is what H1 was designed to measure), and at least one participant who seemed to be using the help button as a strategy to solve the quiz, rather than as a real request for support. Our original study notes actually mention this exact pattern already ('Gaming Behaviour: Peer participants using help systematically to eliminate options'), so this is not a brand-new discovery. But we believe it deserved more attention in this report than just a short note, because it goes a long way toward explaining why H1 came out non-significant, even though the raw average difference looked large. The one strategic participant pulls the peer group's average upward, while several other peer participants who never used the help button at all pull the variation even higher, without moving the average of 'genuine' help-seeking very much.

5.5 Manipulation Check

All fourteen participants answered the manipulation-check question ('Did the robot act more like a classmate or more like a teacher?') in a way that matched their assigned group. All seven peer-group participants described the robot as a classmate. All seven tutor-group participants described it as a teacher. No participants were removed from the study because of this question.

We also want to note that the printed summary produced by our analysis script has a small labelling bug. It shows the tutor group's numbers twice, once for the peer row and once for the tutor row. However, the underlying grouped data confirms that the 100% pass rate reported above is correct.

6 Discussion and Conclusion

Let us return to our research question from Section 1: does a robot's role as a peer, instead of a tutor, change how often students ask for help and how comfortable they feel during learning? Our honest answer is a mixed one, and we think it is more useful to say this plainly, instead of rounding it up to a simple 'yes'.

H2 (comfort) is supported by our numbers. Peer-group participants rated their experience as significantly more comfortable, with a large effect size. H1 (help-seeking) is not supported at the normal significance level, even though peer-group participants asked for help roughly twice as often on average. This is because the peer group’s help-count numbers were extremely varied, and were pushed up strongly by one participant’s strategic use of the help button, rather than by a broad, group-wide change in willingness to ask for support.

More importantly, we do not think the comfort result can be cleanly explained by ‘the robot feeling like a peer instead of an authority figure’, which is exactly what H2 was designed to test. Because the peer group’s hint system gives away more information per request - removing a wrong answer completely, instead of giving a text clue that still needs thinking - and because the peer group also gets an extra retry chance that the tutor group does not, peer-group participants ended up scoring almost twice as high on the quiz. This is a result our study did not predict, and one that, according to our original hypothesis wording (‘even if task scores are similar’), was specifically supposed to not happen. Since doing well on a quiz is, by itself, a reasonable explanation for feeling comfortable afterward, our comfort result is mixed together with our score result. We cannot cleanly separate ‘participants felt more comfortable because the robot felt like a peer’ from ‘participants felt more comfortable because the peer group’s hint system made the quiz much easier’.

This pattern is also worth comparing to Salomons et al. [2022]’s study, which compared peer and tutor robots teaching adults, and found no significant overall difference in learning outcomes between groups. Only participants with low prior knowledge benefited more from the peer robot. Our finding of a large, group-wide score difference does not match well with that result. Together with the option-elimination pattern we found directly in our own help log, this points toward our score gap being caused by our specific hint system design, rather than being a true copy of a real social effect from peer framing.

6.1 Limitations

There are several important limitations in our study.

- **Group assignment, for the data studied in this report, was self-selected, not randomised.** Participants clicked which role they wanted on the starting screen, which does not match the coin-flip randomisation described in both our ethics application and our early project slides. This creates a self-selection bias - for example, participants who were already more comfortable asking for help, or already more relaxed in general, may have been more likely to choose ‘Peer’. A simple between-groups t-test cannot correct for this. Note that the current version of ‘app.py’ now assigns the group randomly on the server (Section 4.2), which fixes this specific problem going forward. However, it does not change how we should interpret the 14 participants’ results reported in this study, since those results were collected before this fix was made.
- **The two hint systems are not equal in the information they give.** The peer hint completely removes a wrong answer and can be used again and again with no limit. The tutor hint only ever gives a fixed text clue. This is very likely the

main reason for the significant score difference in Section 5.2, and it also makes the comfort result in Section 5.2 harder to trust cleanly. This problem still exists in the current code, even after the group-assignment fix above.

- **The spoken message before the quiz (Section 3.2), which is present in the current system, is not balanced between groups.** The peer message is about 45% longer, and it tells participants about the retry chance before the quiz even begins. The tutor message does not mention anything similar. Any future study using the current system should either make both messages equal in length and content, or remove the early mention of the retry chance, since this could shape participants' expectations, and therefore their comfort, in a way that has nothing to do with the peer/tutor framing itself.
- **The peer group gets a second attempt after a first wrong answer.** The tutor group does not. This is another separate reason for the score difference between groups.
- **The way participants answer questions - by voice or by typing - is not equal between groups.** The tutor group has more voice-based questions by default, and the peer group has more typed questions. This adds another variable that is different between groups, but is not related to the peer/tutor framing itself.
- **Our sample size ($n = 14$, 7 per group) is small.** It is below the 20–24 participants planned in our ethics application, and it is likely too small to reliably detect a moderate effect, especially for the very noisy help-count variable, as shown by the non-significant but medium-to-large effect size for H1.
- **At least one participant appears to have used the peer group's help button as a strategy, to remove wrong answers on purpose, instead of using it as a genuine request for support.** This makes it harder to trust DV1 (help requests) as a pure measure of how much support participants actually needed.
- **The manipulation check used only one forced-choice question, with no confidence rating.** Comfort was also measured using only one 1–7 question, instead of a properly validated, multi-question scale.
- **The whole interaction happened through a laptop chat window, not a physically present robot.** Both Belpaeme et al. [2018]'s review and Bainbridge et al. [2011]'s study report that a physically present robot creates stronger social effects. This means our results might be smaller than what a fully physical peer robot would produce, or might be different in other ways too.

6.2 Future Work

A cleaner version of this study, done again in the future, should make the two hint systems equal in the information they provide. For example, both groups could get a single text clue, or both groups could get a single option removed not one of each, mixed across groups as in our study. It should also remove the retry-chance difference, make the two

pre-quiz spoken messages equal in length and content (Section 3.2), and collect the full planned sample of 20–24 participants, with a more balanced mix of genders.

The group-assignment problem is already fixed in the current version of the code ‘app.py’ now assigns the group randomly on the server, instead of relying on participants choosing for themselves. This means a new round of data collection using the current code would no longer have that specific bias, as long as the remaining hint, retry, and message-length problems described above are also fixed first.

It would also be useful to add a limit, or some kind of ‘cost’, to repeated help requests. This way, the help-count variable would measure something closer to a participant’s real, felt need for support, instead of measuring an available strategy for solving the quiz. Given more resources, running the study using a physically present robot, instead of a browser-based chat agent, would also be a natural next step, following the embodiment argument made by Belpaeme et al. [2018]’s review and by Bainbridge et al. [2011]’s study.

Overall, we think this small pilot study adds a small, messy, but honest data point to the gap identified by Belpaeme et al. [2018]’s review where only 9% of all studies involve a robot presented as a peer at all. The biggest lesson we would highlight for anyone building on this work, including ourselves, is not really about the specific numbers. It is about a design trap we fell into. It is very easy, when trying to make a ‘peer’ hint feel supportive and forgiving, to accidentally also make it more informative than the ‘tutor’ hint it is being compared against. Once that happens, any difference in comfort or performance stops being evidence about the social framing of the robot, and instead simply becomes evidence about who was given more of the answer.

Declaration of AI Usage

The following AI-based tools were used for the preparation of this thesis.

Table 3: AI tools used and their purpose

Tool	Purpose of Use
ChatGPT	To correct grammar and spelling mistakes.
Tablesgenerator.com	To create tables and add them to the Overleaf template.
Deep Seek	To Handle with the latex code errors

Reference Links:

- ChatGPT Conversation: <https://chatgpt.com/share/6a7a464c-5a20-83eb-a30d-2536120b42>
- Tables Generator: https://www.tablesgenerator.com/latex_tables

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